

**o-XYLENE****ICSC: 0084 (April 2024)**

ortho-Xylene  
1,2-Dimethylbenzene  
o-Xylo

**CAS #: 95-47-6****UN #: 1307****EC Number: 202-422-2**

|                             | ACUTE HAZARDS                                                      | PREVENTION                                                                                                                                                                                                                                                   | FIRE FIGHTING                                                                                                  |
|-----------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>FIRE &amp; EXPLOSION</b> | Flammable. Above 30°C explosive vapour/air mixtures may be formed. | NO open flames, NO sparks and NO smoking. Above 30°C use a closed system, ventilation and explosion-proof electrical equipment. Prevent build-up of electrostatic charges (e.g., by grounding). NO contact with incompatible materials: See Chemical Dangers | Use water spray, powder, foam, carbon dioxide. In case of fire: keep drums, etc., cool by spraying with water. |

|                   | SYMPTOMS                                                   | PREVENTION                                 | FIRST AID                                                                                                                                                 |
|-------------------|------------------------------------------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b> | Dizziness. Drowsiness. Headache. Nausea.                   | Use local exhaust or breathing protection. | Fresh air, rest. Refer for medical attention.                                                                                                             |
| <b>Skin</b>       | Dry skin. Redness.                                         | Protective gloves.                         | First rinse with plenty of water for at least 15 minutes, then remove contaminated clothes and rinse again. Rinse and then wash skin with water and soap. |
| <b>Eyes</b>       | Redness. Pain.                                             | Wear safety spectacles or face shield.     | First rinse with plenty of water for several minutes (remove contact lenses if easily possible), then refer for medical attention.                        |
| <b>Ingestion</b>  | Burning sensation. Abdominal pain. Further see Inhalation. | Do not eat, drink, or smoke during work.   | Rinse mouth. Do NOT induce vomiting. Refer for medical attention.                                                                                         |

| SPILLAGE DISPOSAL                                                                                                                                                                                                                                                                                                                                                                                                                | CLASSIFICATION & LABELLING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Personal protection: filter respirator for organic gases and vapours adapted to the airborne concentration of the substance.</p> <p>Ventilation. Remove all ignition sources. Do NOT let this chemical enter the environment. Collect leaking and spilled liquid in sealable containers as far as possible. Absorb remaining liquid in sand or inert absorbent. Then store and dispose of according to local regulations.</p> | <p><b>According to UN GHS Criteria</b></p> <div style="text-align: center;">  <p><b>DANGER</b></p> </div> <p>Flammable liquid and vapour<br/>May be fatal if swallowed and enters airways<br/>Harmful in contact with skin or if inhaled<br/>Causes skin irritation<br/>Causes serious eye irritation<br/>May cause damage to central nervous system through prolonged or repeated exposure<br/>Toxic to aquatic life</p> <p><b>Transportation</b><br/><b>UN Classification</b><br/>UN Hazard Class: 3; UN Pack Group: III</p> |
| STORAGE                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Fireproof. Store only in original container. Separated from strong oxidants and strong acids. Store in an area without drain or sewer access.                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| PACKAGING                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |



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**o-XYLENE****ICSC: 0084****PHYSICAL & CHEMICAL INFORMATION****Physical State; Appearance**

COLOURLESS LIQUID WITH CHARACTERISTIC ODOUR.

**Physical dangers**

As a result of flow, agitation, etc., electrostatic charges can be generated.

**Chemical dangers**

Reacts with strong acids and strong oxidants.

Formula:  $C_6H_4(CH_3)_2$  /  $C_8H_{10}$ 

Molecular mass: 106.2

Boiling point: 144°C

Melting point: -25°C

Relative density (water = 1): 0.88 (20°C)

Solubility in water, g/l at 20°C: 0.18 (very slightly soluble)

Vapour pressure, kPa at 20°C: 0.7

Relative vapour density (air = 1): 3.7

Relative density of the vapour/air-mixture at 20°C (air = 1): 1.02

Flash point: 30°C c.c.

Auto-ignition temperature: 465°C

Explosive limits, vol% in air: 0.9-7.6

Octanol/water partition coefficient as log Pow: 3.12

Viscosity: 0.81 mPa\*s at 20°C

Electrical conductivity: 0.067 pS/m

**EXPOSURE & HEALTH EFFECTS****Routes of exposure**

The substance can be absorbed into the body by inhalation, through the skin and by ingestion.

**Effects of short-term exposure**

The substance is irritating to the eyes and skin. If this liquid is swallowed, aspiration into the lungs may result in chemical pneumonitis. The substance may cause effects on the central nervous system. This may result in impaired functions.

**Inhalation risk**

A harmful contamination of the air will be reached rather slowly on evaporation of this substance at 20°C.

**Effects of long-term or repeated exposure**

The substance defats the skin, which may cause dryness or cracking. The substance may have effects on the central nervous system. Animal tests show that this substance possibly causes toxicity to human reproduction or development.

**OCCUPATIONAL EXPOSURE LIMITS**

TLV: 100 ppm as TWA; 150 ppm as STEL; A4 (not classifiable as a human carcinogen); BEI issued.

EU-OEL: 221 mg/m<sup>3</sup>, 50 ppm as TWA; 442 mg/m<sup>3</sup>, 100 ppm as STEL; (skin)**ENVIRONMENT**

The substance is toxic to aquatic organisms.

**NOTES**

Depending on the degree of exposure, periodic medical examination is suggested.

The recommendations on this Card also apply to xylene. Xylene (CAS 1330-20-7) is a technical mixture of the three isomers o-XYLENE (ICSC 0084), m-XYLENE (ICSC 0085), p-XYLENE (ICSC 0086) and ETHYLBENZENE (ICSC 0268).

**ADDITIONAL INFORMATION****EC Classification**

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